

Assignment – 02:

Problem Statement:

Write a menu driven program in C to implement the basic arithmetic operations such as Addition, Subtraction, Multiplication, and Division.

Code:

```
#include <stdio.h>

int main()
{
    char op;
    float num1,num2,result;
    int rem;
    printf("Calculator operations are:\n");
    printf("+ for addition\n");
    printf("- for subtraction\n");
    printf("* for multiplication\n");
    printf("/ for division\n");
    printf("m for modulus\n");
    printf("Enter an operator: ");
    scanf("%c",&op);
    printf("Enter two numbers: ");
    scanf("%f%f",&num1,&num2);
    switch(op)
    {
        case '+': result=num1+num2;
                printf("The sum is: %f",result);
                break;
        case '-': result=num1-num2;
                printf("The difference is: %f",result);
```

```

        break;
case '*': result=num1*num2;
        printf("The product is: %f",result);
        break;
case '/': if (num2!=0)
        {
            result=num1/num2;
            printf("The quotient is: %f",result);
        }
        else
        {
            printf("Error! Division by zero is not defined");
        }
        break;
case 'm': rem=(int)num1%(int)num2;
        printf("The reamainder is: %d",rem);
        break;
default: printf("Sorry, this operator is not defined!");
        break;
}
}

```

Input & Output:

```

Calculator operations are:
+ for addition
- for subtraction
* for multipliclation
/ for division
m for modulus
Enter an operator: +
Enter two numbers: 23 43
The sum is: 66.000000

```

```

Calculator operations are:
+ for addition
- for subtraction
* for multipliclation
/ for division
m for modulus
Enter an operator: -
Enter two numbers: 32 20
The difference is: 12.000000

```

Calculator operations are:

+ for addition

- for subtraction

* for multipliclation

/ for division

m for modulus

Enter an operator: *

Enter two numbers: 21 10

The product is: 210.000000

Calculator operations are:

+ for addition

- for subtraction

* for multipliclation

/ for division

m for modulus

Enter an operator: /

Enter two numbers: 32 2

The quotient is: 16.000000

Calculator operations are:

+ for addition

- for subtraction

* for multipliclation

/ for division

m for modulus

Enter an operator: m

Enter two numbers: 3 2

The reamainder is: 1